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A multimodal framework for enhancing E-commerce information management using vision transformers and large language models

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ABSTRACT

In the rapidly advancing field of visual search technology, traditional methods that rely only on visual features often struggle with accuracy and relevance. This challenge is particularly evident in e-commerce, where precise product recommendations are critical, and is further complicated by keyword stuffing in product descriptions. To address these limitations, this study introduces BiLens, a multimodal recommendation framework that integrates both visual and textual information. BiLens leverages large language models (LLMs) to generate descriptive captions from image queries, which are transformed into word embeddings, and extracts visual features using Vision Transformers (ViT). The visual and textual representations are integrated using an early fusion strategy and compared using cosine similarity, enabling deeper contextual understanding and enhancing the accuracy and relevance of product recommendations in capturing customer intent. A comprehensive evaluation was conducted using Amazon product data across five categories, testing various image captioning models and embedding methods—including BLIP-2, ViT-GPT2, BLIP-Image-Captioning-Large, Florence-2-large, GIT (microsoft/git-base-coco), Word2Vec, GloVe, BERT, and ELMo. The combination of Florence-2-large and BERT emerged as the most effective, achieving a *precision* of 0.81 $\pm$ 0.14 and *F*1 score of 0.49 $\pm$ 0.16. This setup was further validated on the Myntra dataset, showing generalizability with *precision* of 0.59 $\pm$ 0.27, *recall* of 0.47 $\pm$ 0.25, and *F*1 score of 0.52 $\pm$ 0.24. Comparisons with image-only and text-only baselines confirmed the superiority of the fusion-based approach, with statistically significant improvements in F1 scores, underscoring BiLens's ability to deliver more accurate, context-aware product recommendations.

1. Introduction

The exponential growth of e-commerce has significantly reshaped business operations and consumer interactions with products and services. As of 2023, global e-commerce sales reached approximately $6 trillion, with projections indicating a compound annual growth rate (CAGR) of 9.1 % through 2027 (Statista, 2024). This expansion highlights the increasing importance of advanced search functionalities and personalized recommendations in enhancing the online shopping experience and driving conversion rates (Unleashed Technologies, 2024). As the e-commerce landscape continues to evolve, businesses are also recognizing the growing significance of advanced technologies like visual search. The visual search market, valued at $9.21 billion in 2022, is projected to reach $46.19 billion by 2032, while the text analytics market, valued at $9.55 billion in 2023, is expected to grow to $41.2 billion by 2032 (Market Decipher, 2022; SNS Insider Research, 2024). These figures underscore the rapid advancement of both image-based and text-based search methods, signaling the need for businesses to integrate sophisticated search technologies to meet the shifting demands of modern consumers.

While text-based search methods continue to be useful, there is a clear shift towards more intuitive and effective search technologies, especially in the fashion e-commerce sector. Visual search technology is increasingly being adopted to meet the changing needs of consumers. Unlike traditional text search, visual search allows users to upload images to find similar products, offering a more engaging and efficient shopping experience. Studies show that consumers prefer visual search for its convenience and the ability to make their shopping process more enjoyable and straightforward. A study by ViSenze (2023) found that 62 % of millennials and Generation Z are more likely to buy a product when they can search using images. Furthermore, consumers are now looking to replicate the in-store shopping experience online by using visual

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search to capture images of items they see in real life or on social media. This helps them find prices, check availability, and discover more affordable alternatives, often leading to a stronger intent to purchase (Zhang et al., 2018; Bhardwaj et al., 2013; Goel, 2017).

As e-commerce platforms continue to evolve, the increasing demand for visual search and retrieval has driven significant advancements in deep learning (DL) methods, particularly in the areas of image recognition and data retrieval. DL, through complex neural network architectures, automatically extracts intricate features from raw image data, enabling the accurate representation of product visual characteristics (Angadi et al., 2021). This ability to capture detailed visual content is crucial for precise image recognition, categorization, and classification, significantly improving product search and recommendation systems (Shiau et al., 2020). Recent advancements have been further bolstered using transformers, initially developed for natural language processing (NLP), which have shown exceptional promise in computer vision tasks. Vision Transformers, for example, have revolutionized feature extraction and image classification, leading to more effective and efficient downstream applications. Additionally, the application of transfer learning and pre-trained models, which leverage large datasets, has proven instrumental in enhancing e-commerce image recognition. This approach accelerates training processes and boosts model accuracy with minimal data, ultimately improving product identification and recommendation precision (Samia et al., 2022). Large Language Models sophisticated deep learning models that excel at NLP and generative AI tasks, have been crucial in providing contextual understanding of product images. These models, trained on vast datasets with transformer architectures, generate human-like text, creating accurate image captions that enrich search and recommendation systems (Zhao et al., 2023). By integrating these models, e-commerce platforms enhance user experience, allowing for more accurate and efficient product searches that go beyond the limitations of traditional text-based search methods (Gupta & Shete, 2024; Zhang et al., 2022). This synergy of deep learning techniques not only improves operational efficiency but also transforms the way consumers interact with online retail platforms, making product discovery more intuitive and effective.

Visual search technology has proven beneficial across various domains, including healthcare, education, and the food industry. However, the most anticipated and widely adopted use of visual search technology is in e-commerce, where it significantly enhances the shopping experience by enabling image-based product searches. This approach not only increases user engagement but also correlates with higher clicks, likes, and purchases, as visual relevance, such as color, texture, and design, strongly influences user behavior (Togashi & Sakai, 2020). By incorporating visual search, e-commerce platforms replicate an offline shopping experience, where consumers rely on visual inspection, such as color and patterns, to make purchase decisions. As noted by Bhardwaj et al. (2013), this method adds a layer of intuitive discovery to online retail, making product search feel more natural and user-friendly. Leading e-commerce sites, such as Amazon, have already integrated visual search to enhance product discovery, driving substantial growth. Amazon, for instance, achieved nearly $575 billion in net sales revenue worldwide in 2023, with 6.1 billion direct visits to Amazon.com as of December 2023 (Statista, 2024).

In e-commerce, visual search engines are designed to improve the accuracy and relevance of image-based product searches. These systems typically consist of key components such as image preprocessing, feature extraction, and search algorithms. While these components provide significant advantages, image-only search methods still face several limitations. These systems mainly focus on basic visual attributes, such as color, shape, and background, which creates a gap between the primitive features they detect and the more complex, higher-level concepts users intend. As a result, search outcomes may include products that superficially resemble the desired item but fail to align with the user's actual needs. For instance, a visual search query for a black t-shirt may return results for items with a black background or similar colors, rather than the intended garment (Goodrum & Spink, 2001).

To overcome these challenges, multimodal visual search has emerged as an effective solution, combining both visual and textual data, which makes use of product titles and descriptions, to provide a more accurate representation of the user's query (Bendouch et al., 2023). However, even this approach presents its own set of issues, particularly when it comes to product metadata. A common problem in traditional text-based search is keyword stuffing, where sellers overload product titles with excessive or irrelevant keywords in an attempt to improve visibility. This practice results in cumbersome and misleading product titles, such as "S925 Sterling Silver Marquise Shape Jade Aventurine Open Leaf Ring Olive Leaf Ring Handmade Jewelry Unique Gifts for Women Mother Mom Wife" (Ni et al., 2019). These inflated titles not only degrade the accuracy of text-based search but also affect multimodal search systems. When visual search queries are combined with these verbose product titles, the inclusion of irrelevant or overly broad keywords can lead to inaccurate results, as the search algorithm may prioritize the textual metadata over the visual content. For example, a visual search query for a specific handbag might return irrelevant results if the title includes excessive keywords like "Fashionable Casual Shoulder Handbag Leather Bag Large for Women Men Gifts." Keyword stuffing leads to mismatches between the user's intent and the results, as multimodal search systems still struggle to strike the right balance between text and visual data. Furthermore, text-based systems often fail to capture intricate visual details such as patterns or textures, which makes it difficult for both image-only and multimodal search systems to deliver precise and relevant results, ultimately complicating the user experience (Beall, 2008).

To address the limitations of traditional visual search methods, this study explores the use of LLMs to enhance visual search by automatically adding contextual understanding to user visual queries. Traditional image search methods struggle to accurately interpret user intent, and issues like keyword stuffing and misspellings in product descriptions further hinder search precision. By integrating LLM-generated captions into a multimodal framework, we aim to improve both the accuracy of visual search and product recommendations. Unlike existing methods that rely solely on visual features or metadata, this approach utilizes LLMs to generate context-rich captions for the image query. These captions act as a natural language description of the image, akin to how a user would phrase a search query. As a result, the system processes visual queries with the same precision as traditional text-based searches, eliminating the need for users to manually input keywords, while also incorporating the image's features into the query for enhanced accuracy. Our research specifically examines the following questions:

RQ1: How do large LLMs enhance the effectiveness of visual search frameworks in e-commerce platforms?
RQ2: In what ways do different LLMs vary in generating captions for e-commerce images, and how do these variations affect search accuracy?
RQ3: How does incorporating generated captions into image search methodologies overcome the limitations of traditional image search approaches?
RQ4: How can image search technologies help resolve the challenges associated with product titles in text-based search systems within e-commerce?
RQ5: What impact does the choice of word embedding techniques have on the performance of image search in e-commerce applications?

While numerous studies have explored visual search methodologies, the application of LLMs for image captioning in e-commerce image search remains limited. This study aims to address this gap by proposing an end-to-end recommendation framework, BiLens, which utilizes LLMs to generate captions for consumer-uploaded images. These captions serve as supplementary input, offering a semantic representation of the user's query to the recommendation engine. This approach addresses the challenge of recommendation engines that primarily focus on visual features while often neglecting detailed product attributes. A popular metric for measuring similarity in high-dimensional spaces is cosine similarity, which has been widely used in text-based applications, such as evaluating semantic alignment between student-written scripts and GPT-generated embeddings (Himi et al., 2024). By leveraging this metric, our proposed system extends its application to multimodal embeddings, combining both visual and textual features. Cosine similarity enables effective comparisons of image-text pairs, allowing for the precise retrieval of relevant products based on a consumer's image query. Vision Transformers (ViT) are used to extract high-level visual features through self-attention mechanisms, while various word embedding techniques has been experimented enhance the semantic content of the generated captions. The study will evaluate three approaches, with the first two serving as baselines:

- **Image-only**: The engine uses only the features extracted from the uploaded image to make recommendations. This approach serves as a baseline for visual-based search.
- **Text-only**: The recommendation engine uses only the generated captions as the semantic representation of the user's image query, essentially performing a keyword search based on the image's captions instead of relying on product metadata.
- **Multimodal/Fusion**: The proposed solution integrates both the image query and the generated captions, combining visual and textual data for a more comprehensive and accurate search result.

By comparing these methods, the study aims to evaluate how combining visual and textual data enhances recommendation accuracy. We hypothesize that BiLens will improve customer satisfaction and conversion rates by delivering more precise product recommendations and overcoming the limitations of traditional image search methods. This, in turn, will enhance user experience and boost business performance. The research has the potential to transform how consumers interact with e-commerce platforms, making online shopping more intuitive and efficient by providing accurate, personalized product suggestions. Furthermore, it introduces a novel approach for e-commerce retrieval systems to better understand user input, mimicking human-like comprehension through advanced natural language techniques. This enables more effective and contextually relevant product recommendations. The key contributions of this paper include:

- **Amazon Dataset**: The multimodal approach proposed, integrating BERT with Florence-2-large, achieves an F1 score of 0.81. This shows a significant improvement over text-only methods, which achieve an F1 score of 0.28, and far exceeds the performance of image-only approaches, which have an F1 score of 0.12.
- **Myntra Dataset**: The multimodal approach proposed achieves an F1 score of 0.52, outperforming both the text-only method (0.30) and the image-only method (0.28), demonstrating significant improvements in performance with the multimodal integration.
- Presents an in-depth analysis of various image captioning and LLM integration techniques, alongside different word embedding methods, establishing a new benchmark for enhancing visual search performance through a multimodal approach.
- Showcases BiLens capacity to boost efficiency and customer satisfaction in the e-commerce sector, illustrating the practical use of LLMs by meeting contemporary consumer demands and providing actionable strategies for businesses, with statistically significant improvements over traditional unimodal approaches.

2. Related works

2.1. Advancements in large language models (LLMs) for E-commerce

The rapid development of large language models (LLMs) has brought about a significant transformation in natural language processing (NLP), facilitating substantial advancements across various domains (Khurana et al., 2022; Zhao et al., 2023). Models such as BERT (Devlin et al., 2019), Open Pretrained Transformer (OPT) (Zhang et al., 2022), and the GPT series (Radford et al., 2019) have set new standards for essential NLP tasks, including text classification, generation, summarization, and question answering. The transformer architecture, the backbone of these models, has been pivotal in achieving remarkable accuracy and coherence in language processing (Vaswani et al., 2023). These innovations have revolutionized the processing and utilization of language, driving innovation in various sectors, particularly e-commerce.

In e-commerce, LLMs have proven to be essential in enhancing multiple aspects of the shopping and customer experience. They contribute to predictive analytics for personalized search and recommendations, improve sentiment analysis by synthesizing product reviews, and enable generative analytics for creating product information, all while streamlining customer service with advanced question-answering capabilities (Peng et al., 2024; Ren et al., 2024). Notable models such as E-BERT, KG-FLIP, RecMind, and EComGPT have demonstrated the powerful impact of LLMs on e-commerce. E-BERT, for example, fine-tuned on the Amazon Dataset, has significantly improved customer interactions by enhancing sentiment analysis accuracy in product-related tasks (Zhang et al., 2021). KG-FLIP, by incorporating knowledge graphs, improves contextual understanding, thus enhancing product search functionality (Jia et al., 2023). EComGPT, trained on the EcomInstruct dataset, excels in zero-shot learning for tasks like attribute extraction and product title generation, showcasing its versatility in tackling various e-commerce challenges (Li et al., 2023). Similarly, LLaMA-E refines LLaMA for specialized applications such as ad generation and product QA, demonstrating cutting-edge results in zero-shot performance (Shi et al., 2023). RecMind enhances recommender systems by interpreting user behavior more accurately, providing tailored recommendations for users (Wang et al., 2024). These advancements highlight the growing trend of AI integration within e-commerce, with empirical evidence underscoring the potential of LLMs to drive sales and boost customer loyalty.

2.2. Image recognition and captioning

Image recognition and captioning have become critical in integrating computer vision and natural language processing, particularly for applications like e-commerce. Deep learning approaches such as the Neural Image Captioning (NIC) model (Vinyals et al., 2015) and more recent transformer-based architectures have enabled the generation of semantically rich and syntactically coherent descriptions (Hossain, Sohel, Shiratuddin, & Laga, 2019). Models like CLIP (Radford et al., 2021) and BLIP-2 (Li et al., 2023) have significantly advanced Vision-and-Language tasks by better aligning visual content with textual representations, while large-scale pre-trained models demonstrate improved generalization across domains (Shen et al., 2021). These advancements are particularly impactful in e-commerce, where accurate and context-aware product descriptions enhance product discovery and customer engagement. Recent research has further improved fashion-specific captioning using attention-enhanced encoder-decoder architectures (Nguyen et al., 2021), and ensemble-based caption generation using large language models has been shown to produce more human-like outputs (Bianco et al., 2023).

2.3. Visual search in E-Commerce

Research into visual search technologies has demonstrated their transformative potential across various domains. Visual search technologies have revolutionized product discovery in e-commerce by enabling users to search using images instead of text queries. Major fashion e-commerce platforms like Myntra and Amazon have integrated these capabilities to improve user engagement and shopping experience. Visual features such as color, texture, and design are critical in fashion product search and recommendation, directly influencing user behavior, including clicks and purchases (Togashi & Sakai, 2020). Advances in deep learning have driven the development of scalable and accurate visual search systems tailored to e-commerce fashion. For example, Yang et al. (2017) proposed a deep neural network with binary hashing trained on large eBay image collections, providing fast and precise retrieval. Li et al. (2019) developed a real-time visual search system for JD.com, capable of handling hundreds of billions of fashion product images with low latency and high throughput.

In fashion-specific applications, Kim et al. (2017) designed a large-scale product search framework combining fashion-attribute recognition with visual feature extraction using ResCeption-LSTM and Faster R-CNN. Their approach achieved high precision and recall in attribute recognition and competitive retrieval performance on a dataset of millions of fashion-product images. Additionally, Angadi et al. (2021) improved online shopping recommender systems by combining CNN-based similarity detection and transfer learning, achieving notable accuracy gains on fashion datasets.

These developments demonstrate how visual search enhances fashion e-commerce by enabling more context-aware and user-centric product discovery. Continued innovation in this area is essential for meeting the demands of dynamic fashion markets and improving customer satisfaction.

2.4. Multimodal search in E-commerce

Multimodal search in e-commerce enhances user experience by integrating visual and textual features, improving search accuracy and recommendation quality. Laenen et al. (2018) demonstrated substantial gains in the fashion domain using a multimodal retrieval model on the Amazon Dresses dataset, achieving improvements of 267 % in precision@1, 158 % in precision@5, and 253 % in MAP over baselines. Tang et al. (2024) advanced multimodal retrieval with pre-trained models like instructBLIP and CLIP to generate image-based captions fused with text descriptions, achieving an NDCG score of 0.75 on the Amazon ECI dataset, outperforming both image-only and text-only approaches. Bhattacharya et al. (2019) developed a multimodal dialog system that combines text and image queries within a joint embedding space, yielding high cosine similarity in visual browsing tasks. Han et al. (2017) fine-tuned GoogleNet on image-text data for fashion-attribute recognition, obtaining a V-measure score of 0.658 on the Fashion200K dataset, surpassing traditional models. Zoghbi et al. (2016) introduced a cross-modal search system for fashion products, achieving a precision of 48.75 % and recall of 43.66 % at $k = 5$, significantly outperforming prior methods. These studies collectively highlight the critical role of multimodal approaches in fashion e-commerce, demonstrating clear improvements in recommendation precision, recall, and overall user satisfaction.

3. Methodology

3.1. Proposed framework architecture

This study introduces BiLens, a novel product recommendation framework designed to enhance precision by utilizing product image analysis. BiLens retrieves relevant product recommendations based on a given input image, leveraging advanced visual analysis and contextual understanding to improve the e-commerce search experience. By ensuring that users receive pertinent product recommendations based on their image queries, BiLens effectively handles user-uploaded images and integrates multiple modalities to generate highly relevant recommendations.

The framework operates through an online workflow that incorporates several key stages, each crucial for efficient image processing, feature extraction, and recommendation generation. The workflow, as depicted in Fig. 1, consists of the following steps:

1. **User Image Upload:** The process begins with the user uploading an image, which serves as the primary input for the search process. The quality and clarity of the uploaded image are critical, as they directly influence the image processing and feature extraction stages that follow.
2. **Image Captioning:** After the image is uploaded, an image captioning LLM processes the image, performing essential preprocessing tasks such as resizing and normalizing pixel values, internally. The LLM generates concise captions that provide a textual representation of the image's content. This step is crucial, as it converts visual data into text, making it compatible with subsequent analysis. The length and detail of the captions are adjustable via a hyperparameter, allowing for flexibility in the depth of information provided.
3. **Feature Extraction:** In this phase, features are extracted from the uploaded image using a pretrained vision transformer model, as shown in Fig. 2. The model captures high-dimensional attributes of the image, including textures, shapes, and colors. Simultaneously, text embeddings are generated from the captions using a word embedding model such as BERT. These embeddings represent the semantic meaning of the captions in a vector space, facilitating comparison and retrieval. Feature extraction is vital for converting raw image and text data into structured, analyzable formats.
4. **Feature Combination and Fusion Analysis:** At this stage, we combine the features extracted from the uploaded image with the textual features derived from the generated captions. The fusion technique integrates these two data types—visual content and text—into a unified feature vector. This multimodal combination

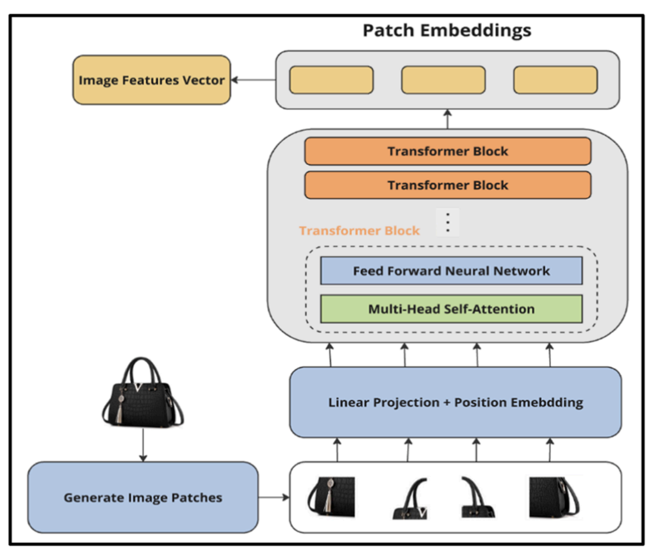

Fig. 2. Proposed BiLens Framework Overview An illustration of the six-stage BiLens framework for online recommendations: [1] User Input Flow, [2] Feature Extraction by ViT, [3] Image Captioning by LLM, [4] Feature Extraction by Word Embedding Techniques, [5] Fusion of Image and Caption Vectors, and [6] Recommendation Logic. This figure outlines the entire process, from capturing user input to generating personalized recommendations, culminating in the display of the top-recommended similar products. Additionally, the figure includes baseline approaches for comparison: [1, 2, 3, 4] Image-Only and [1, 2, 3, 4, 5] Text-Only methods, highlighting the advantages of the proposed framework.

allows the framework to leverage both the visual characteristics of the image (e.g., textures, colors, and shapes) and the semantic content captured in the caption. By merging these features, the framework gains a richer, more comprehensive understanding of the input, which enhances its ability to generate more relevant product recommendations. To assess the effectiveness of this fusion approach, we compare it with two baseline methods: the image-only approach, which uses only the visual features of the uploaded image, and the text-only approach, which relies on captions generated from the uploaded image rather than product titles. These baseline methods are depicted in Fig. 1.

5. **Recommendation Logic:** In the final stage, the framework computes cosine similarity between the fused features of the consumer-uploaded image and those of products in the database, utilizing both image and caption vectors. This similarity measure helps in identifying and retrieving the top *k* products most relevant to the input image. The value of *k* is determined through experimentation with various feature combinations, ensuring that the search results are precise and closely aligned with the user's query, thereby enhancing the overall user experience.

3.2. Data collection and preprocessing

The dataset utilized for this study was derived from the Amazon Review Data (2018) collection (Ni et al., 2019), which encompasses a wide range of features, including both review data and product metadata. For the purposes of this research, we focused on specific product metadata elements, namely the asin (a unique product identifier), title (the product name), and imageURLHighRes (the URL of the high-resolution product image). The asin was essential for uniquely identifying each product within the dataset, while the title was employed for comparison with the captions generated by large language models (LLMs), as product titles are reliable, succinct descriptors directly provided by sellers. The imageURLHighRes feature was crucial to ensure high-quality, clear images for effective search results.

Other features, such as the product description and imageURL (associated with lower resolution images), were excluded from the analysis. The description field often contained detailed, product-specific information that, while comprehensive, was deemed irrelevant to the scope of this framework, as it tended to be overly elaborate for the purpose of generating concise search queries. For instance, the product description in Table 1 provides extensive detail about a watch's design, materials, and features, which is valuable for some use cases but not essential for our search system. In contrast, the product title serves as an efficient and practical identifier, which aligns with the observation that 56 % of retail site search queries consist of a single word (Statistica, 2022). Furthermore, studies such as Zoghbi et al. (2013) report that the average query length in Amazon datasets is approximately 8.06 words, reinforcing the relevance of using product titles in search systems. The dataset contained both low and high-quality images, yet only high-resolution images were utilized. This decision was made to

Table 1
Sample Product Data - This table compares lengthy product descriptions with their corresponding short product titles, illustrating the concise summarization of product details.

Product Image	Product Description	Product Title
	Uncommon valor from the Officer's Series…Swiss Army women's watch in high performance stainless steel. Dynamic, contemporary lines are casual and relaxed. Comfortable brushed steel case is fully integrated into the brushed and polished steel bracelet. Polished steel bezel frames the charcoal gray dial with luminous hands and Arabic numeral hour markers. A window at the 3 o'clock hour reveals the automatic date calendar. Striking fluted steel crown. Bracelet has a push-button clasp for added convenience. Scratch-resistant sapphire crystal. Water resistant to 30 m (100 feet). Precise Swiss quartz movement. Case is 22.5 mm wide and 8 mm thick.	Swiss Army Officer_Watch Watch 024,598

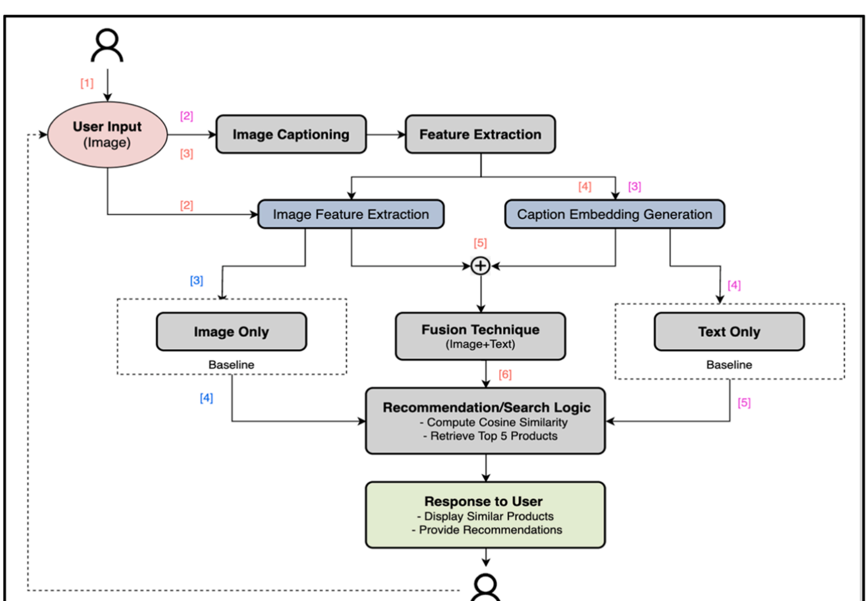

Fig. 1. Vision Transformer for Feature Extraction An illustration of the Vision Transformer (ViT) architecture used for feature extraction, featuring a product image of a handbag from the Amazon dataset employed in this study. The figure details the model's components, including input image patch embeddings, self-attention layers, and the resulting feature outputs.

emphasize how intricate image features influence relevant suggestions, ultimately improving the accuracy and reliability of the search system in this experiment.

The Amazon dataset consists of 21 distinct product categories. For this research, we selected five categories—Clothing, Shoes & Sandals, Jewelry, Bags & Wallets, and Accessories—to serve as use cases for our proposed framework. This selection was also driven by computational resource limitations, which necessitated a more focused analysis. A total of 1000 images were randomly sampled to ensure a representative subset, while also optimizing for computational efficiency. Although this study primarily focused on these five categories, the framework is designed to be extensible, enabling the inclusion of additional categories as needed. This flexibility is essential, especially in the context of visual search technology, which has seen widespread adoption for image-based searches in sectors like fashion and accessories. For example, Alibaba's "search by image" application, which had over 17 million daily active users in 2017 (Zhang et al., 2018), underscores the growing importance of visual search in e-commerce.

Preprocessing is a crucial step in preparing data for effective and accurate analysis. The preprocessing workflow includes image validation and downloading, where images are checked for accessibility and functionality using the imageURLHighRes feature before being downloaded to Google Drive for storage. Following this, image resizing adjusts the images to a standard 224 × 224 pixels, ensuring compatibility with the input requirements of the Vision Transformer (ViT) and maintaining consistency across the dataset (Dosovitskiy et al., 2021). This resizing step is essential for enabling consistent feature extraction from the images. In addition, text preprocessing involved cleaning the product titles by removing special characters, punctuation marks, and converting the text to lowercase, ensuring consistency, and eliminating discrepancies caused by formatting variations. After the initial preprocessing steps, which included handling missing values and filtering records, 7996 unique products were identified in the dataset. From this, a random selection of 1000 images was made to balance the diversity of products with the available computational resources. This selection allowed for a manageable yet representative dataset for evaluating the framework's performance. A sample of the randomly selected product images is shown in Fig. 3.

To further evaluate the recommendation results of our proposed framework, we introduced a new column titled "similar items" in the dataset. This column was manually curated to include related products based on references from Amazon's "Related Products" section for the 1000 randomly selected records. As shown in Fig. 4, this column provides the "similar items" for each product in the dataset, ensuring a more comprehensive evaluation of the framework's recommendations. The accuracy and relevance of these entries were verified by three independent members, who cross-checked the related products for consistency. This verification process was essential for ensuring the integrity of the dataset and enabling reliable evaluation of the recommendation system's performance.

In addition to the Amazon dataset, we used the Myntra dataset for testing purposes, focusing on the same five fashion categories. This addition to the dataset provided a wider range of data for model validation and testing. The Myntra dataset also enabled us to assess the generalizability of our framework across different sources and contexts within the fashion domain.

3.3. Analysis and model selection

3.3.1. Image captioning models

Image captioning adds essential context when consumers upload product images, helping the system better interpret visual inputs. To achieve this, we applied image captioning models leveraging pre-trained LLMs and transformer architectures for their strong generalization and performance (Gupta & Shete, 2024).

In this study, we explored five state-of-the-art models, each utilizing advanced multimodal learning approaches. These models were selected for their robustness, scalability, and adaptability in e-commerce scenarios, where accurate, context-aware captions are critical for enhancing user experience and recommendation effectiveness. These models were pretrained on large-scale datasets covering a wide range of real-world objects—such as clothing, footwear, bags, and accessories—allowing them to produce high-quality, context-rich captions. This, in turn, significantly improves recommendation precision by aligning product attributes with consumer queries. The variation in their captioning strategies—from fine-grained visual descriptions to broader contextual narratives—further enhances the system's capability to deliver relevant and meaningful outputs.

1. **BLIP-2:** Employs a CLIP-like image encoder, a Querying Transformer (Q-Former), and an OPT-2.7B LLM. It is fine-tuned on the LAION dataset, bridging the gap between visual and textual embeddings to generate detailed, context-aware captions (Li, Li, Savarese & Hoi, 2023).
2. **ViT-GPT2:** Combines Vision Transformer (ViT) for extracting visual features with GPT-2 for caption generation. Fine-tuned on the Flickr8k dataset, it leverages strong self-attention mechanisms to generate coherent and relevant text outputs (Mishra et al., 2024).
3. **BLIP-Image-Captioning-Large:** Features a ViT-based encoder and a unified multimodal encoder-decoder (MED). Trained on synthetic and filtered captions and fine-tuned on the COCO dataset, it excels at generating accurate, domain-relevant captions (Li, Li, Xiong & Hoi, 2022).
4. **Florence-2-large:** A unified sequence-to-sequence model that combines a ViT encoder and a transformer-based multimodal decoder. Fine-tuned on the FLD-5B dataset, it supports diverse vision-language tasks with strong generalization capabilities (Xiao et al., 2023).
5. **GIT (microsoft/git-base-coco):** Utilizes a contrastively pre-trained image encoder and a transformer decoder, fine-tuned on the COCO

Fig. 3. Sample of Product Images - This figure displays a set of randomly selected product images used in the experimentation process to evaluate the effectiveness of the proposed method.

Fig. 4. Example of Similar Items An illustration showing the "similar items" column populated with related product suggestions for a given men's short. The figure displays five different products like the input shorts, demonstrating how related items are curated and presented. The examples are sourced from the Amazon dataset used in the study.

dataset. It is optimized for both image captioning and broader vision-language understanding tasks (Wang et al., 2022).

Table 2 provides an overview of the architectural components, including the image encoders, language models used, and the number of parameters for each model.

3.3.2. Word embedding generation

For preprocessing product image captions in the dataset, we explored various Natural Language Processing (NLP) techniques to generate word embeddings, capturing both semantic and syntactic relationships. These word embeddings help in transforming textual data into numerical formats suitable for multimodal input. The techniques employed are as follows:

1. **GloVe** was utilized for its ability to generate dense word embeddings by analyzing global word co-occurrence statistics, capturing semantic relationships effectively (Pennington et al., 2014).
2. **Word2Vec** was employed for its efficiency in learning word embeddings through context prediction, offering context-sensitive vectors that capture word usage nuances (Jatnika et al., 2019).
3. **ELMo** was chosen for its contextualized embeddings, which provide richer word representations based on surrounding text, enhancing understanding of word roles in various contexts (Peters et al., 2018).
4. **BERT** was selected for its advanced bidirectional context understanding, producing nuanced, context-aware embeddings by analyzing both preceding and following text (Devlin et al., 2019).

3.3.3. Image feature extraction

For preprocessing product images, we selected the ViT-B/16 model with a Masked Autoencoder (MAE) approach, owing to its advanced transformer architecture, which excels in visual feature extraction. This model employs a transformer-based encoder to process visible image patches and a lightweight decoder to reconstruct masked regions from latent representations. Trained on the ImageNet-1 K dataset, this approach improves both efficiency and accuracy, making it highly suitable for generating robust image embeddings that enhance search and retrieval tasks (He et al., 2022).

The preprocessing pipeline included model-specific normalization and resizing transforms. Visual features were extracted using PyTorch and stored in NumPy format for subsequent analysis in the multimodal learning pipeline.

Table 2
Architecture Details of Image Captioning LLMs - This table outlines the architecture details of various image captioning language models selected for experimentation, including their image encoders, language models or encoders, and parameter counts.

Model	Image Encoder	Language model/ encoder	#Param.
BLIP-2 (2023)	ViT-g/14	OPT-2.7b	3.8B
ViT-GPT2 (2022)	ViT-B/16	GPT-2	302M
BLIP-Image-Captioning-Large (2022)	ViT	Transformer (MED)	470M
Florence-2-large (2023)	DaViT	Transformer	0.77B
GIT (2022)	DaViT	Transformer	5.1B

3.3.4. Fusion technique

Multimodal input plays a critical role in improving e-commerce search and recommendation tasks by combining product images with their associated captions to offer a more complete understanding of product context. In this study, Early Fusion was adopted as it merges image and text features into a single unified representation. This approach enables the model to learn interactions between the two modalities from the beginning, resulting in more accurate and context-aware outputs.

Initially, each modality is processed independently to create unimodal embeddings. Text embeddings are generated using techniques such as GloVe, Word2Vec, ELMo, and BERT. Image features are extracted using the ViT-B/16 model with a masked autoencoder. These features are then integrated through concatenation, forming a cohesive multimodal representation.

The dimensionality of the final vector depends on the text embedding used. The combination of ViT-B/16 and BERT produces a 1536-dimensional vector. ViT-B/16 paired with GloVe or Word2Vec results in a 1068-dimensional vector, while pairing with ELMo produces a 1792-dimensional vector, as shown in Fig. 5. These unified vectors strengthen the model's ability to capture both visual and textual information, enhancing its effectiveness in search and recommendation systems.

3.3.5. Search/Recommendation logic

Cosine similarity was selected for identifying similar products based on image and text features due to its ability to compare high-dimensional vectors by evaluating their orientation rather than their magnitude. This method is well-suited for analyzing complex embeddings, focusing on the relative orientation of feature vectors, which enhances accuracy in matching products based on their content and context. Retrieved products will be ranked by cosine similarity scores, ensuring that the most relevant product images are presented to the user.

$$\textit{Cosine Similarity} = (F1 \bullet F2) \,/\, (|F1| * |F2|) \tag{1}$$

where $F1$ and $F2$ are the fused feature vectors that combine both image and text features for two items being compared. Specifically, $F1$ represents the feature vector for the input query image, integrating both visual and caption data. $F2$ represents the feature vector for each product image in the database, also incorporating visual and caption features. The term $F1 \cdot F2$ denotes the dot product of these vectors, which measures the alignment between them. The magnitudes $|F1|$ and $|F2|$ are the lengths of the feature vectors in the multi-dimensional space.

3.4. Experimental setup

To assess the performance of various image captioning LLMs, we utilized a diverse set of pretrained models recognized for their ability to generate accurate and contextually rich captions. These models, built on

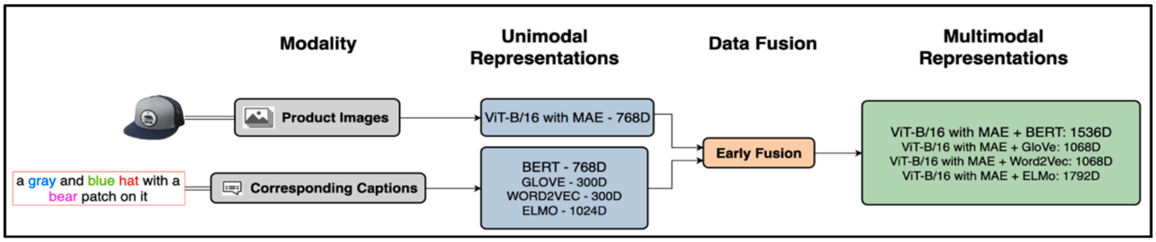

Fig. 5. Overview of Modality Integration and Representations An illustration showing the stages of integrating different data modalities (captions and images) into unified representations. It includes the processing of each modality to generate embeddings, followed by early fusion to create a cohesive vector, and the resulting multimodal representations with varying dimensions.

transformer architectures and trained on benchmark datasets, were applied to a curated set of 1000 product images to construct multimodal vectors for our product catalog.

The evaluation process considered key hyperparameters, including batch size, beam width (which determines the number of beams used in beam search), and the minimum and maximum caption lengths to control the output range. Table 3 summarizes the hyperparameters configured for each image captioning model used in our experiments. Inference was conducted primarily on CUDA-enabled GPUs for computational efficiency, with CPU-based processing available as a fallback to ensure flexibility.

The batch size and beam width were selected based on the model size and the available GPU capabilities. Larger models, such as BLIP2-OPT-2.7B, Microsoft GIT-Base-COCO, and Florence-2-Large, were able to support higher batch sizes and beam widths due to their higher memory requirements and efficient handling on A100 GPUs. Mid-sized models like ViT-GPT2-Image-Captioning and BLIP-Image-Captioning-Large were configured with moderate values to ensure smooth and optimized performance without overloading the system. The minimum and maximum caption lengths were kept consistent across all models to maintain uniformity in generated outputs.

All product images were resized to 224 × 224 pixels to match the input size expected by the ViT model, and the classification head was removed to extract feature representations instead.

All computations were performed using a combination of local and cloud resources. The local environment included an Apple M2 chip with 8GB RAM and 4GB graphics memory, while cloud-based processing used A100 GPUs for faster computation.

The workflow utilized a range of libraries and frameworks: PyTorch and Torchvision for deep learning and model inference; TensorFlow and TensorFlow Hub for additional support; Hugging Face Transformers for accessing pretrained models; PIL and Timm for image processing; NumPy, Pandas, and Scikit-learn for numerical and data manipulation; Gensim, NLTK, and JSON for text processing; and Matplotlib for visualizations.

Table 3
Hyperparameters for Image Captioning LLMs - This table details the hyperparameters for different image captioning LLMs used in the experiment, including batch size, beam amount, minimum caption length, and maximum caption length.

Model Name	Batch Size	Beam Amount	Min caption Length	Max caption Length
blip2-opt-2.7b	10	7	15	30
vit-gpt2-image-captioning	8	5	15	30
blip-image-captioning-large	8	5	15	30
Florence-2-large	10	7	15	30
microsoft_git-base-coco	10	7	15	30

3.5. Evaluation metrics

3.5.1. Evaluation metrics for image captioning

To evaluate image captioning within the framework, we focus on assessing the performance of the model to ensure that the generated captions are accurate and semantically rich, as they are crucial for determining similarity based on both images and their descriptions. This evaluation involves comparing the generated captions against product titles in the dataset to ensure alignment and relevance. High-quality captions enhance the matching process for consumer-uploaded images, improving the effectiveness of product recommendations and the overall search experience. We assume that high-quality captions contribute significantly to this enhancement by providing more accurate and contextually relevant descriptions, thus aligning better with user queries and product attributes.

We assess the image captioning model's performance using Bilingual Evaluation Understudy (*BLEU*) and Metric for Evaluation of Translation with Explicit Ordering (*METEOR*) scores.

BLEU Score: It evaluates the precision of generated captions by analyzing the overlap of *n-grams* (*1-grams, 2-grams, 3-grams, and 4-grams*) between the generated captions and reference product titles, capturing various aspects of text similarity (Papineni et al., 2022).

$$BLEU = BP * exp((1/N) * \Sigma (w_n * log(p_n))) \tag{2}$$

$$\mathrm{BP} = \{1, \text{ if } c > r \exp(1 - r / \mathrm{c}), \text{ if } c \le r\} \tag{3}$$

$$Pn = \frac{No.\ of\ n - grams\ in\ candidate\ that\ appear\ in\ references}{Total\ no.\ of\ n - grams\ in\ candidates} \tag{4}$$

In Eq. (2), ***Brevity Penalty (BP),*** w_n are the weights for different n-grams, $log(p_n)$ is the logarithm of the n-gram precision. The *BP* is used to penalize captions that are shorter than the reference product titles. In Eq. (3), *c* represents the length of the candidate (generated) caption, and *r* represents the length of the reference product titles. In Eq. (4), precision *Pn* for n-grams is calculated as the ratio of the number of *n-grams* in the candidate translation that are also in the reference translations to the total number of n-grams in the candidate translation. The *BLEU* score uses the geometric mean of *n-gram* precision scores to provide a balanced evaluation that emphasizes the exact matches and word order in the generated captions, ensuring they align closely with human-written reference titles provided by sellers.

METEOR score: It evaluates the quality of generated captions by considering multiple factors such as precision, recall, stemming, synonyms, and paraphrases. It provides a comprehensive assessment by capturing semantic similarity and handling variations in wording (Banerjee & Lavie, 2005).

$$METEOR = F\_mean \times (1 - Penalty) \tag{5}$$

$$F\_mean = \frac{10 \times (Precision \times Recall)}{(Recall + 9 \times Precision} \tag{6}$$

In Eq. (5), *F_mean* is the harmonic mean of *Precision* and *Recall*, and the *Penalty* is applied to account for the alignment between generated and reference texts. In Eq. (6), *Precision* measures the proportion of *n-grams* in the generated caption that match those in the reference titles. *Recall* measures the proportion of *n-grams* from the reference titles that are included in the generated caption. The *Penalty* term is a factor that reduces the score if the generated caption's alignment with reference title is poor, considering issues such as incorrect word order or lack of fluency. The ***METEOR*** score thus combines these elements to evaluate how well the generated captions align with human-written reference product titles, offering a balanced measure of both exact and semantic accuracy.

3.5.2. Evaluation metrics for recommendation results

To assess the effectiveness of the BiLens search framework, we use *Precision@k*, Recall@k, and *F1@k* metrics, where *k* is the number of top-ranked search results considered when evaluating the performance of a search or recommendation system (Krasnov, 2024).

Precision@k: It measures the proportion of relevant items among the top-*k* retrieved results. Eq. (7) evaluates the accuracy of the search results by checking how many of the top-*k* items are actually relevant.

$$Precision@k = \frac{Number\ of\ Relevant\ Items\ in\ Top - k}{k} \tag{7}$$

Recall@k: It measures the proportion of relevant items that are retrieved within the top-*k* results. Eq. (8) assesses the completeness of the search results by indicating how many of the total relevant items were found in the top-*k* results.

$$Recall@k = \frac{(Number\ of\ Relevant\ Items\ in\ Top - k)}{(Total\ Number\ of\ Relevant\ Items)} \tag{8}$$

F1@k: It is the harmonic mean of *Precision@k* and *Recall@k*. Eq. (9) provides a single metric that balances both precision and recall, offering a comprehensive measure of search result quality.

$$F1@k = \frac{2 * (Precision@k * Recall@k)}{(Precision@k + Recall@k)} \tag{9}$$

To ensure that observed differences between models are not due to chance, we also perform statistical significance testing using the Wilcoxon signed-rank test. This non-parametric test is well-suited for comparing paired metric values (such as F1 scores) across different models or configurations when the assumption of normality may not hold (Wilcoxon, 1945). A p-value <0.05 is considered statistically significant, indicating strong evidence that one model outperforms another.

4. Experimental results

4.1. Evaluation of image captioning LLMs

To evaluate the performance of various image captioning LLMs, we used five different randomly selected sample images, as shown in Table 4. Table 4 includes the product image, title, and captions generated by each image captioning LLMs: Florence-2-large (M1), BLIP2-opt-2.7b (M2), BLIP-image-captioning-large (M3), Microsoft_GIT-base-COCO (M4), and ViT-GPT2-image-captioning (M5). It also includes the corresponding BLEU and METEOR scores for the generated captions, which are standard metrics used to evaluate the quality of captions in comparison to the product titles.

Table 4 presents the performance metrics for five different sample images using various image captioning LLMs across different product categories. For Image 1, the *BLEU* scores ranged from 0.01 to 0.03, and the *METEOR* scores ranged from 0.05 to 0.15. Overall, M2 performed best with a high *METEOR* score of 0.15 and a competitive *BLEU* score, while M4 performed the worst with the lowest scores in both metrics. For Image 2, the *BLEU* scores ranged from 0.01 to 0.02, and the *METEOR* scores ranged from 0.16 to 0.18. M1 and M5 performed the best, both

Table 4
Performance evaluation results of image captioning LLMs.

Product image	Title	Generated Caption	*BLEU* Score	*METEOR* Score
Image 1: 	Simple Pleated Handbag Satin Evening Prom Clutch Bridal Bag Purse	M1: a black clutch bag with a chain strap	0.03	0.10
		M2: a black satin clutch bag with a chain strap and a silver chain	0.02	0.15
		M3: a black satin clutch bag with a chain strap and a clasp on the front	0.02	0.14
		M4: a small black leather purse with a chain on the side	0.01	0.05
		M5: a black and white photo of a white purse	0.02	0.05
Image 2: 	GUCCI SUNGLASSES	M1: a pair of sunglasses on top of a table	0.02	0.18
		M2: an image of a pair of sunglasses with a brown frame and brown lenses	0.01	0.16
		M3: there is a close up of a pair of sunglasses on a white surface	0.01	0.16
		M4: a pair of sunglasses with a pair of sunglasses on a white surface	0.01	0.16
		M5: a pair of sunglasses sitting on top of a table	0.02	0.18
Image 3: 	rugged brown sandal	M1: a pair of brown sandals on a white background	0.02	0.52
		M2: a pair of women's leather sandals with a buckle on the heel	0.0	0.12

(*continued on next page*)

Table 4 (*continued*)

Product image	Title	Generated Caption	*BLEU* Score	*METEOR* Score
		M3: pair of women's brown leather wedged sandals with buckles on each side	0.01	0.25
		M4: a pair of women's shoes with a design on the sole	0.0	0.0
		M5: a pair of black leather shoes sitting on top of a table	0.0	0.0
Image 4: 	Snowflake Necklace for Women Sterling Silver Cubic Zirconia Snowflake Pendant with Adjustable 18–20-inch Chain, Christmas Jewelry for Teen Girls Mom Girlfriend	M1: a white gold snowflake pendant With diamonds on a chain	0.03	0.17
		M2: a snowflake shaped pendant with diamonds on a white gold plated chain	0.02	0.14
		M3: a close up of a necklace with a snowflake design on it	0.01	0.07
		M4: a black and white photo of a brooch in the shape of a flower	0.0	0.02
		M5: a decorative vase with a bird on top of it	0.0	0.02
Image 5: 	Men's Shorts Casual Classic Fit Drawstring Summer Beach Shorts with Elastic Waist and Pockets	M1: men's casual shorts with drawstring waistband	0.03	0.35
		M2: men's shorts with zippers on the side and drawstring on the waist	0.09	0.39
		M3: a close up of a pair of shorts with zippers on the side	0.02	0.13
		M4: i'm in love with this pair of men's leggings	0.03	0.17
		M5: a black and white photo of a black and white shoe	0.01	0.03

Individual BLEU and METEOR scores for five randomly selected product images, showing titles and captions generated by Florence-2-large [M1], BLIP2-opt-2.7b [M2], BLIP-image-captioning-large [M3], Microsoft_GIT-base-COCO [M4], and ViT-GPT2-image-captioning [M5]. Overall, [M1], followed by [M2], provides the best results in caption quality.

achieving the highest *METEOR* score of 0.18 and tied for the highest BLEU score of 0.02, while M2, M3, and M4 had slightly lower performance with a *METEOR* score of 0.16 and a *BLEU* score of 0.01. For Image 3, the *BLEU* scores ranged from 0.0 to 0.02, and the *METEOR* scores ranged from 0.0 to 0.52. M1 performed the best with the highest *METEOR* score of 0.52 and a *BLEU* score of 0.02, while M4 and M5 were the worst performers, both scoring 0.0 in both *BLEU* and *METEOR*. For Image 4, the *BLEU* scores ranged from 0.0 to 0.03, and the *METEOR* scores ranged from 0.02 to 0.17. M1 achieved the highest *BLEU* score of 0.03 and the highest *METEOR* score of 0.17, indicating the best performance. In contrast, M4 and M5 had the lowest scores, both recording 0.0 in *BLEU* and 0.02 in *METEOR*. For Image 5, the *BLEU* scores ranged from 0.01 to 0.09, and the *METEOR* scores ranged from 0.03 to 0.39. M2 achieved the highest *BLEU* score of 0.09 and the highest *METEOR* score of 0.39, indicating the best performance. Conversely, M5 had the lowest scores, with a *BLEU* score of 0.01 and a *METEOR* score of 0.03.

In this evaluation, *METEOR* scores consistently surpassed *BLEU* scores for each model, highlighting their superior ability to assess caption relevance. The evaluation results of various image captioning LLMs from Table 4, revealed distinct performance patterns. The M1 model achieved a highest *METEOR* score of 0.52 for Image 3, demonstrating its capability to generate relevant captions in terms of meaning and context. However, it generally had lower *BLEU* scores, with a maximum of 0.03, indicating less precision in matching the exact wording of the product titles. M2 emerged as the top performer, attaining the highest *BLEU* score of 0.09 and *METEOR* score of 0.39 for Image 5. This model consistently produced captions with higher accuracy and relevance across multiple images. The M3 model showed solid performance, particularly for Img 3 with a *METEOR* score of 0.25 but had lower *BLEU* scores, peaking at 0.02 across all images. M4 performed best for Img 2 with a *BLEU* score of 0.10 and *METEOR* score of 0.16, but its results were otherwise limited, suggesting restricted versatility. Finally, M5 consistently scored the lowest, with a maximum *BLEU* score of 0.02 and *METEOR* score of 0.18, indicating it struggled to generate high-quality, descriptive captions across all images. Overall, BLIP2-opt-2.7b stands out as the most effective model in this comparison, while the other models show varying degrees of performance limitations.

Table 5 summarizes the overall performance of various image captioning LLMs, presenting mean and standard deviation values for

Table 5
Performance Evaluation Results for Caption Generation LLMs. *Mean* and *Standard Deviation* of *BLEU* and *METEOR* scores for Caption Generation LLMs: among all five LLMs, Florence-2-large outperforms the others.

Model name	*BLEU Score*	*METEOR Score*
blip2-opt-2.7b	0.0271 ± 0.0263	0.1499 ± 0.1185
vit-gpt2-image-captioning	0.0062 ± 0.0099	0.0229 ± 0.0376
blip-image-captioning-large	0.0202 ± 0.0196	0.1137 ± 0.1022
Florence-2-large	0.0343 ± 0.0566	0.1452 ± 0.1411
microsoft_git-base-coco	0.0149 ± 0.0140	*0.787* ± 0.0858

BLEU and *METEOR* scores across a dataset of 1,000 records. The mean scores reflect the average performance across all samples, while the standard deviation highlights the variability in scores for each model. The BLIP2-opt-2.7b model stands out as the most effective, with an average *BLEU* score of 0.0271 ± 0.0263 and a leading *METEOR* score of 0.1499 ± 0.1185. This model excels in producing captions that are both contextually accurate and relevant, demonstrating a well-rounded performance. In comparison, the Florence-2-large model achieves the highest *BLEU* score of 0.0343 ± 0.0566 but has a slightly lower *METEOR* score of 0.1452 ± 0.1411. This indicates a stronger ability to match exact wording with product titles, though it is less effective in capturing contextual nuances. The BLIP-image-captioning-large model, with an average *BLEU* score of 0.0202 ± 0.0196 and a *METEOR* score of 0.1137 ± 0.1022, shows significant performance in contextual relevance but with lower precision in exact matches. The Microsoft_GIT-base-COCO model records a *BLEU* score of 0.0149 ± 0.0140 and a *METEOR* score of 0.0787 ± 0.0858, reflecting moderate effectiveness with lower consistency. Lastly, the ViT-GPT2-image-captioning model exhibits the lowest scores, with a *BLEU* score of 0.0062 ± 0.0099 and a *METEOR* score of 0.0229 ± 0.0376, indicating challenges in generating high-quality captions. Overall, Florence-2-large demonstrates the most balanced performance across both metrics, showing strong potential for generating accurate and contextually relevant captions, which can be valuable for enhancing product descriptions and search functionalities in e-commerce platforms.

4.2. Evaluation of recommendation results

To assess the performance of our BiLens search framework, we employed *Precision@k*, *Recall@k*, and *F1@k* as key evaluation metrics. The framework was tested across three feature combinations: an image-only model, a text-only model, and a fusion model integrating both image and text. Here, the parameter k represents the number of top predictions considered in calculating the *Precision*, *Recall*, and *F1 scores*. We evaluated the framework with $n = 100$ input queries, allowing us to compute the mean and standard deviation for *precision*, *recall*, and *F1 scores*. The image-only model used the uploaded query image as input, while the text-only model generated captions for the uploaded image, utilizing these captions for the search. The fusion model incorporated both the uploaded image and the generated text captions. This approach was adopted to assess the effectiveness of various feature combinations against the base image-only and text-only approaches. Our experiments included 20 model combinations for each feature type, combining four different word embedding models with five distinct image captioning LLMs. Thus, we evaluated 20 combinations for the text-only approach, 1 for image-only, and 20 for the fusion approach. To identify the optimal hyperparameter k, the models were evaluated across values from 10 to 30 in increments of 5 (i.e., 10, 15, 20, 25, and 30). Considering the trade-off between precision, recall, and F1 scores for each feature combination, $k = 10$ was selected as the most appropriate value for subsequent experiments.

4.2.1. Experimental evaluation results of image-only approach

In this study, we initially evaluated the performance of the BiLens search framework using the image-only feature combination. This approach relies solely on image vectors extracted by the ViT model to match consumer-uploaded images with product images in the dataset. As presented in Table 6, the evaluation focuses exclusively on visual features, without leveraging any textual information or additional context. At $k = 10$, across 100 input queries, the image-only model achieved a mean Precision of 0.44 (±0.25), a mean Recall of 0.08 (±0.05), and a mean F1 score of 0.12 (±0.07). These results indicate that while the ViT-based image representation provides a reasonable basis for visual similarity matching, its limited recall and moderate precision highlight challenges in distinguishing visually similar but contextually distinct products. The lack of textual context constrains the model's ability to accurately interpret and rank relevant products. This image-only performance establishes a baseline for comparison with subsequent experiments incorporating text features, such as image captions, and with the fusion approach that integrates both image and text representations.

Table 6
Performance evaluation results for image-only recommendations using ViT at $k = 10$ this serves as a baseline.

Model	Feature Combination	*Precision@*10	*Recall@*10	*F1@*10
ViT	Image Only	0.44 ± 0.25	0.08 ± 0.05	12 ± 0.07

4.2.2. Experimental evaluation results of text-only approach

The evaluation of text-only feature combinations for the BiLens, utilized captions generated for consumer-uploaded images and matched them with product image captions in the database. Various text embeddings, including BERT, GLOVE, Word2Vec, and ELMO, were applied to these captions to assess the effectiveness of different text embedding techniques in retrieving relevant product images. The performance of each combination is detailed in Table 7, which includes 20 combinations of 5 image captioning LLMs paired with 4 types of word embeddings.

The performance of image captioning models with text-only feature combinations revealed substantial differences in effectiveness. Florence-2-large with Word2Vec embedding achieved the highest performance, with a mean precision@10 of 0.67 and a standard deviation of 0.18, a

Table 7
Performance evaluation results for text-only feature combinations at $k = 10$ – The Florence-2-large + Word2Vec combination performed better, highlighting the effectiveness of Word2Vec in text-only feature sets.

Model	Word Embedding	*Precision@*10	*Recall@*10	*F1@*10
Florence-2-large	BERT	0.62 ± 0.21	0.35 ± 0.16	0.41 ± 0.12
Florence-2-large	GLOVE	0.59 ± 0.20	0.34 ± 0.16	0.39 ± 0.12
Florence-2-large	**Word2Vec**	0.67 ± 0.18	0.39 ± 0.18	0.45 ± 0.11
Florence-2-large	ELMO	0.59 ± 0.20	0.34 ± 0.17	0.39 ± 0.12
blip2-opt-2.7b	BERT	0.49 ± 0.21	0.27 ± 0.14	0.32 ± 0.12
blip2-opt-2.7b	GLOVE	0.45 ± 0.21	0.26 ± 0.17	0.30 ± 0.13
blip2-opt-2.7b	**Word2Vec**	0.53 ± 0.20	0.31 ± 0.19	0.35 ± 0.13
blip2-opt-2.7b	ELMO	0.43 ±0.20	0.25 ± 0.16	0.28 ± 0.12
blip-image-captioning-large	BERT	0.41 ±0.19	0.22 ± 0.11	0.26 ± 0.11
blip-image-captioning-large	GLOVE	0.40 ± 0.19	0.22 ± 0.11	0.26 ± 0.10
blip-image-captioning-large	**Word2Vec**	0.50 ± 0.20	0.28 ± 0.17	0.33 ± 0.14
blip-image-captioning-large	ELMO	0.35 ± 0.18	0.19 ± 0.11	0.23 ± 0.11
microsoft_git-base-coco	BERT	0.35 ± 0.20	0.18 ± 0.12	0.22 ± 0.10
microsoft_git-base-coco	GLOVE	0.32 ± 0.18	0.17 ± 0.11	0.20 ± 0.09
microsoft_git-base-coco	**Word2Vec**	0.37 ± 0.19	0.19 ± 0.11	0.23 ± 0.09
microsoft_git-base-coco	ELMO	0.32 ± 0.19	0.17 ± 0.10	0.20 ± 0.09
vit-gpt2-image-captioning	BERT	0.23 ± 0.16	0.12 ± 0.09	0.15 ± 0.10
vit-gpt2-image-captioning	GLOVE	0.23 ± 0.17	0.13 ± 0.09	0.15 ± 0.11
vit-gpt2-image-captioning	**Word2Vec**	0.25 ± 0.19	0.14 ± 0.10	0.17 ± 0.12
vit-gpt2-image-captioning	ELMO	0.23 ± 0.17	0.13 ± 0.09	0.15 ± 0.11

mean recall@10 of 0.39 and a standard deviation of 0.18, and a mean F1@10 score of 0.45 and a standard deviation of 0.11. This model's advanced vision foundation capabilities and comprehensive dataset usage contributed to its superior performance. In contrast, the blip2-opt-2.7b model with Word2Vec embedding showed competitive results, with a mean precision@10 of 0.53 and a standard deviation of 0.20, a mean recall@10 of 0.31 and a standard deviation of 0.19, and a mean F1@10 score of 0.35 and a standard deviation of 0.13. The blip-image-captioning-large model followed, achieving a mean precision@10 of 0.50 and a standard deviation of 0.20, a mean recall@10 of 0.28 and a standard deviation of 0.17, and a mean F1@10 score of 0.33 and a standard deviation of 0.14. Microsoft_git-base-coco demonstrated lower performance, with a mean precision@10 of 0.37 and a standard deviation of 0.19, a mean recall@10 of 0.19 and a standard deviation of 0.11, and a mean F1@10 score of 0.23 and a standard deviation of 0.09. The vit-gpt2-image-captioning model exhibited the lowest performance, with a mean precision@10 of 0.25 and a standard deviation of 0.19, a mean recall@10 of 0.14 and a standard deviation of 0.10, and a mean F1@10 score of 0.17 and a standard deviation of 0.12. The progressive decline from Florence-2-large to vit-gpt2-image-captioning indicates decreasing effectiveness in caption generation and image content understanding, with Florence-2-large emerging as the most effective model in terms of caption quality and its impact on search accuracy. This addresses RQ2 by highlighting how different LLM models vary in generating captions for e-commerce images and their effects on search accuracy.

In the text-only approach, the Word2Vec embedding method consistently outperformed other word embedding techniques across various image captioning LLMs, as shown in Table 7. For Florence-2-large, Word2Vec achieved the highest mean precision@10 of 0.67 and a standard deviation of 0.18, a mean recall@10 of 0.39 and a standard deviation of 0.18, and a mean F1@10 score of 0.45 and a standard deviation of 0.11. Word2Vec's effectiveness in capturing semantic similarities between terms contributed to its superior performance. In comparison, the BERT embedding achieved a mean precision@10 of 0.62 and a standard deviation of 0.21, a mean recall@10 of 0.35 and a standard deviation of 0.16, and a mean F1@10 score of 0.41 and a standard deviation of 0.12, demonstrating its ability to capture contextual information, though not as effectively as Word2Vec. GLOVE and ELMO embeddings exhibited slightly inferior results, with GLOVE providing a mean precision@10 of 0.59 and a standard deviation of 0.20 and a mean F1@10 score of 0.39 and a standard deviation of 0.11, and ELMO showing a mean precision@10 of 0.59 and a standard deviation of 0.20 and a mean F1@10 score of 0.39 and a standard deviation of 0.12. ELMO performed the lowest among the embeddings used with Florence-2-large, reflecting its less effective performance in capturing contextual nuances compared to other embeddings. These findings address RQ5 by demonstrating how different word embedding techniques impact image search performance in e-commerce applications, revealing significant variations in effectiveness.

Comparing text-only combinations with image-only feature combinations highlights a substantial improvement when incorporating textual information. The best-performing text-only configuration, Florence-2-large with Word2Vec embedding, achieved a mean precision@10 of 0.67 and a standard deviation of 0.18, a mean recall@10 of 0.39 and a standard deviation of 0.18, and a mean F1@10 score of 0.45 and a standard deviation of 0.11. In contrast, the image-only feature combination using the ViT model resulted in a mean precision@10 of 0.29 and a standard deviation of 0.22, a mean recall@10 of 0.14 and a standard deviation of 0.07, and a mean F1@10 score of 0.17 and a standard deviation of 0.08. This improvement demonstrates that incorporating generated captions provides a richer contextual understanding, significantly enhancing search accuracy and relevance. This effectively addresses RQ3 by showing how the inclusion of generated captions in image search methodologies overcomes the limitations of traditional image-only search approaches.

The evaluation results indicate that the Florence-2-large model with Word2Vec embedding is the most effective approach for BiLens, demonstrating superior performance in text-based matching and significantly improving search accuracy.

4.2.3. Experimental evaluation results of fusion approach

The evaluation of image and text fusion feature combinations for BiLens, involved integrating captions generated from consumer-uploaded images with image vectors extracted using the ViT model. This approach aimed to enhance the matching accuracy between consumer-uploaded images and product images in the dataset. To assess the effectiveness of different text embedding techniques in this context, captions were processed using BERT, GLOVE, Word2Vec, and ELMO embeddings. The results, detailed in Table 8, encompass 20 combinations of five image captioning LLMs with four text embeddings.

The performance of image captioning models using the fusion approach revealed significant differences in effectiveness. Florence-2-large with BERT embedding achieved the highest metrics, with a mean precision@10 of 0.81 and a standard deviation of 0.14, a mean recall@10 of 0.37 and a standard deviation of 0.15, and a mean F1 score@10 of 0.49 and a standard deviation of 0.16, reflecting its advanced vision capabilities and effective integration with BERT's contextual understanding for accurate image representation. The blip2-opt-2.7b model with BERT embedding achieved a mean precision@10 of 0.65 and a standard deviation of 0.18, a mean recall@10 of 0.36 and a standard deviation of 0.15, and a mean F1 score@10 of 0.45 and a standard deviation of 0.16, showing strong captioning capabilities but

Table 8
Performance Evaluation Results for Fusion (Image + Text) Approach at $k = 10$ – The Florence-2-large + BERT combination achieved the best performance, emphasizing the effectiveness of **BERT** in fusion approaches.

Model	Word Embedding	*Precision*@10	*Recall*@10	*F1*@10
Florence-2-large	**BERT**	0.81 ± 0.14	0.37 ± 0.15	0.49 ± 0.16
Florence-2-large	GLOVE	0.62 ± 0.24	0.28 ± 0.13	0.37 ± 0.17
Florence-2-large	Word2Vec	0.54 ± 0.26	0.26 ± 0.15	0.34 ± 0.18
Florence-2-large	ELMO	0.78 ± 0.15	0.37 ± 0.14	0.49 ± 0.16
blip2-opt-2.7b	**BERT**	0.65 ± 0.18	0.36 ± 0.15	0.45 ± 0.16
blip2-opt-2.7b	GLOVE	0.52 ± 0.24	0.26 ± 0.14	0.33 ± 0.17
blip2-opt-2.7b	Word2Vec	0.48 ± 0.25	0.24 ± 0.15	0.31 ± 0.18
blip2-opt-2.7b	ELMO	0.59 ± 0.21	0.32 ± 0.14	0.40 ± 0.16
blip-image-captioning-large	**BERT**	0.57 ± 0.21	0.30 ± 0.14	0.37 ± 0.15
blip-image-captioning-large	GLOVE	0.47 ± 0.23	0.25 ± 0.13	0.31 ± 0.16
blip-image-captioning-large	Word2Vec	0.46 ± 0.24	0.23 ± 0.14	0.30 ± 0.17
blip-image-captioning-large	ELMO	0.54 ± 0.22	0.28 ± 0.12	0.35 ± 0.14
microsoft_git-base-coco	**BERT**	0.52 ± 0.24	0.29 ± 0.17	0.36 ± 0.19
microsoft_git-base-coco	GLOVE	0.46 ± 0.24	0.25 ± 0.15	0.31 ± 0.18
microsoft_git-base-coco	Word2Vec	0.47 ± 0.26	0.24 ± 0.15	0.30 ± 0.18
microsoft_git-base-coco	ELMO	0.48 ± 0.25	0.27 ± 0.17	0.32 ± 0.19
vit-gpt2-image-captioning	BERT	0.42 ± 0.21	0.22 ± 0.10	0.27 ± 0.12
vit-gpt2-image-captioning	GLOVE	0.46 ± 0.21	0.23 ± 0.13	0.30 ± 0.15
vit-gpt2-image-captioning	**Word2Vec**	0.46 ± 0.23	0.23 ± 0.14	0.29 ± 0.16
vit-gpt2-image-captioning	ELMO	0.42 ± 0.22	0.22 ± 0.09	0.27 ± 0.12

lower performance compared to Florence-2-large. The blip-image-captioning-large model with BERT embedding recorded a mean precision@10 of 0.57 and a standard deviation of 0.21, a mean recall@10 of 0.30 and a standard deviation of 0.14, and a mean F1 score@10 of 0.37 and a standard deviation of 0.15, indicating competent results yet not matching the top performers. The microsoft_git-base-coco model with BERT embedding achieved a mean precision@10 of 0.52 and a standard deviation of 0.24, a mean recall@10 of 0.29 and a standard deviation of 0.17, and a mean F1 score@10 of 0.36 and a standard deviation of 0.19. The vit-gpt2-image-captioning model with BERT embedding produced the lowest performance, with a mean precision@10 of 0.42 and a standard deviation of 0.21, a mean recall@10 of 0.22 and a standard deviation of 0.10, and a mean F1 score@10 of 0.27 and a standard deviation of 0.12, due to less effective caption generation and lower semantic alignment with image content. These results demonstrate the varying effectiveness of image captioning models, with Florence-2-large emerging as the most effective when paired with BERT embedding.

For word embedding techniques, the BERT embedding method consistently demonstrated superior performance across various image captioning LLMs, as outlined in Table 3. Specifically, for the Florence-2-large + ViT model, BERT achieved the highest metrics, with a mean precision@10 of 0.81, a mean recall@10 of 0.37, and a mean F1@10 of 0.49. In contrast, the ELMO embedding showed comparable results, attaining a mean precision@10 of 0.78, a mean recall@10 of 0.37, and a mean F1@10 of 0.49, though it exhibited marginally lower precision and recall compared to BERT. GLOVE, while effective, presented lower performance with a mean precision@10 of 0.62, a mean recall@10 of 0.28, and a mean F1@10 of 0.37. Word2Vec, on the other hand, recorded the lowest scores among the embeddings, with a mean precision@10 of 0.54, a mean recall@10 of 0.26, and a mean F1@10 of 0.34. These results underscore BERT's dominance in embedding techniques for the Florence-2-large + ViT model, highlighting its superior ability to enhance image captioning performance. The consistent advantage of BERT was also observed across other image captioning models as shown in Table 8, reinforcing its overall effectiveness in improving performance with the fusion approach.

The integration of text and image features demonstrated a significant enhancement in performance compared to using image-only or text-only approaches. As illustrated in Table 8, the Florence-2-large model utilizing the BERT embedding with the image + text feature combination achieved superior results. Specifically, the model yielded a mean precision@10 of 0.81 and a standard deviation of 0.14, a mean recall@10 of 0.37 and a standard deviation of 0.15, and a mean F1 score@10 of 0.49 and a standard deviation of 0.16. This performance markedly exceeded that of the image-only feature combination with the ViT model, which reported a mean precision@10 of 0.44 and a standard deviation of 0.25, a mean recall@10 of 0.08, and a standard deviation of 0.05, and a mean F1 score@10 of 0.12 and a standard deviation of 0.07. Further examination of text-only approaches using the Florence-2-large model with Word2Vec embedding revealed a mean precision@10 of 0.67 and a standard deviation of 0.18, a mean recall@10 of 0.39 and a standard deviation of 0.18, and a mean F1 score@10 of 0.45 and a standard deviation of 0.11. Although these results are notable, they are still less effective compared to the image + text combination. These findings underscore the substantial benefit of integrating textual information with image features. The incorporation of text provides additional contextual information, thereby improving the relevance and accuracy of search results.

4.2.4. *Quantitative evaluation of baseline and proposed approaches with top performing models*

The following section presents a comprehensive comparison between the approach introduced in this study and existing methods in the field. Existing research in the field of multimodal recommendation systems has demonstrated the prevalence of hybrid models that rely on both image and text inputs, with users typically expected to provide both modalities. For instance, Tautkute et al. (2019) introduced a system that integrates both image and text inputs to facilitate fashion and interior design searches. In contrast, the approach proposed in this study specifically targets the fashion sector and leverages LLMs to automatically generate human-like captions for user-uploaded fashion images. To the best of our knowledge, this represents the first information retrieval system to utilize LLM-generated captions from user-uploaded images, focusing on the product while filtering out background details. Unlike traditional systems, which depend on the user to provide both image and text inputs, this method generates captions directly from the image itself, thus reducing the need for manual text entry. By minimizing reliance on user-provided textual input, the proposed method offers a more intuitive and seamless search experience, addressing existing gaps in hybrid recommendation models. This innovation streamlines the image search process for consumers by automatically generating captions that capture user intent, leading to more accurate and relevant product recommendations without requiring manual textual input.

We selected the best-performing models from our experimental evaluation for the overall comparison, as outlined in the above sections. Specifically, BERT for word encoding, Vision Transformer for image encoding, and Florence-2-large for image captioning demonstrated strong performance across the respective tasks. To further assess the robustness and effectiveness of our proposed fusion-based recommendation approach, we conducted experiments on two fashion-oriented e-commerce datasets: Amazon Fashion and Myntra. Although both datasets belong to the fashion domain, they differ in visual aesthetics, product presentation styles, and textual description patterns, thereby offering diverse input distributions for evaluating the model's adaptability.

We compared three recommendation strategies: Image-Only, Text-Only, and Fusion, where Image-Only and Text-Only serve as unimodal baselines, using standard evaluation metrics: Precision@10, Recall@10, and F1@10. While the unimodal Image-Only and Text-Only approaches exhibited competitive performance in certain metrics, the Fusion strategy consistently outperformed both by leveraging complementary signals from both modalities. On the Amazon Fashion dataset, the fusion approach significantly improved Recall@10 (0.37) and F1@10 (0.49), while maintaining a high Precision@10 (0.81), as shown in Table 9. Similarly, on the Myntra dataset, the fusion method achieved the best overall performance with a Precision@10 of 0.59, Recall@10 of 0.47, and F1@10 of 0.52, as shown in Table 10.

Given the importance of precision and recall in fashion e-commerce, where users demand relevant and diverse recommendations, these results underscore the versatility, generalizability, and real-world applicability of the proposed fusion model within multimodal recommendation systems. Table 11 reports the statistical significance of the proposed fusion model compared to the Image-only and Text-only baselines across both the Myntra and Amazon Fashion datasets, achieved employing nonparametric Wilcoxon Signed-Rank Test. In all cases, the fusion model demonstrates a statistically significant improvement in F1 score, providing strong evidence of its superior performance at the 0.05 significance level.

4.2.5. *Qualitative evaluation of baseline and proposed approaches with top-performing models*

To further support the above quantitative results, a qualitative

Table 9
Performance Evaluation for Amazon Fashion Dataset at $k = 10$ – Fusion approach outperforms Image-Only and Text-Only methods, achieving the highest F1-score.

Approach	*Precision*@10	*Recall*@10	*F1*@10
Image-Only	0.44 ± 0.25	0.08 ± 0.05	0.12 ± 0.07
Text-Only	0.86 ± 0.13	0.18 ± 0.13	0.28 ± 0.14
Fusion	0.81 ± 0.14	0.37 ± 0.15	0.49 ± 0.16

Table 10
Performance Evaluation for Myntra Dataset at $k = 10$ – Fusion approach outperforms Image-Only and Text-Only methods, achieving the highest F1-score.

Approach	*Precision*@10	*Recall*@10	*F1*@10
Image-Only	0.49 ± 0.31	0.20 ± 0.14	0.28 ± 0.19
Text-Only	0.54 ± 0.28	0.22 ± 0.13	0.30 ± 0.16
Fusion	0.59 ± 0.27	0.47 ± 0.25	0.52 ± 0.24

Table 11
Statistical Significance of Fusion Compared to Other Approaches ($k = 10$).

Dataset	Models	Statistical Significance
Myntra	**Fusion** vs Image-Only	1.98×10^{-8}*
	Fusion vs Text-Only	8.06×10^{-3}*
Amazon Fashion	**Fusion** vs Image-Only	4.60×10^{-17}*
	Fusion vs Text-Only	3.25×10^{-15}*

* Statistical significance considering 0.05 significance level.

comparison of the recommendations from the Image-Only, Text-Only, and Fusion approaches was conducted across different fashion categories in both the Amazon and Myntra datasets, validating their effectiveness in delivering relevant and tailored product suggestions for visual product recommendation, where the input is an image.

Table 12 presents the results for the Amazon Fashion dataset. In the Image-Only approach, the recommended products differ significantly from the input product. The model emphasizes visual cues such as black and silver colors while missing the actual context of the item. This often leads to irrelevant matches, showing the limitations of relying solely on image features without semantic guidance. In the Text-Only approach, most of the recommended products are relevant, except for instances where visual details are essential. Although the input product image is a black leather keychain with a silver carabiner, one of the recommended products is a brown keychain. This highlights how purely textual input can miss subtle visual distinctions that are crucial in fashion recommendation. In contrast, the Fusion approach, our proposed multimodal strategy, demonstrates superior performance. As presented in Table 11, all recommended products exhibit a close alignment with the input image in both visual appearance and contextual relevance. The integration of image captions further enhances the model's understanding, improving the accuracy of image-based product recommendations through the combination of visual and textual information.

Table 13 presents the results for the Myntra dataset. In the Image-Only approach, the model successfully recommends some relevant products but also shows inconsistencies. For example, although the input image depicts a navy-blue formal shirt, the recommended products include a casual shirt and an irrelevant black shirt. This suggests that the Image-Only approach may emphasize certain visual features, such as color, but fails to capture the full context of the product, leading to mismatches. In the Text-Only approach, most recommendations align with the input image, except for the last suggestion, which shows a green formal shirt that is not relevant to the input. This highlights how relying solely on text may miss important visual characteristics that are crucial for accurate fashion recommendations. In contrast, the Fusion approach, our proposed multimodal strategy, outperforms the other methods. As presented in Table 13, all recommended products align closely with the input image in terms of both appearance and context. By incorporating image captions, the model leverages both visual and textual information, improving the accuracy and relevance of the product recommendations.

4.2.6. *Performance evaluation and optimization opportunities*

All models employed in the BiLens system, namely Vision Transformers for image embedding generation, Florence-2 for image captioning, and BERT for caption embedding, were subjected to dynamic quantization to enhance computational efficiency. The quantized models were serialized and subsequently loaded during inference to minimize memory footprint and improve runtime performance.

To assess the system's practical applicability, the end-to-end latency for processing a user query and generating personalized product recommendations was systematically evaluated. The total execution time was measured to be approximately 5.36 s, with individual stages contributing as follows: 3.25 s for image embedding generation, 1.00 s for image captioning, 0.70 s for caption embedding, and 0.41 s for recommendation retrieval.

While the observed latency exceeds the thresholds typically expected for real-time deployment, it underscores valuable opportunities for system-level optimization. Future enhancements may include the application of model distillation techniques, advanced quantization strategies (e.g., INT8 quantization), model pruning, embedding caching mechanisms, and the integration of asynchronous processing pipelines. Additionally, leveraging high-performance inference engines such as

Table 12
Recommendation Samples from Amazon Dataset – fusion approach provides the most relevant product recommendations.

Approach	Recommendations
Image-Only	
Text-Only	
Fusion	

Table 13
Recommendation Samples from Myntra Dataset – Fusion approach provides the most relevant product recommendations.

Approach	Recommendations
Image-Only	
Text-Only	
Fusion	

TensorRT and deploying on specialized hardware accelerators (e.g., GPUs or TPUs) could significantly reduce inference time and support real-time responsiveness.

5. Discussion

5.1. Contribution to the E-commerce industry

BiLens introduces a transformative approach to visual search in the e-commerce industry by leveraging AI to improve key operations such as product matching, personalized recommendations, and search optimization. This advancement is crucial, given that analysis of click-through data shows consumers primarily engage with top-ranked recommendations. By improving accuracy in these top-ranked results within the visual search framework, BiLens increases relevance and user engagement, thereby boosting the effectiveness of e-commerce recommendation engines. BiLens addresses the need for better recommendation systems by integrating user-uploaded images with contextual captions. This integration streamlines product discovery and enhances search accuracy, providing e-commerce businesses with an effective solution for improving their recommendation systems and overall search effectiveness.

The integration of visual and textual data significantly enhances recommendation precision and boosts conversion rates, as customers are more likely to purchase products that align closely with their interests (Zumstein & Kotowski, 2020). By delivering more relevant search results, BiLens reduces bounce rates and helps users find suitable products more quickly and efficiently. The system's ability to provide highly personalized suggestions creates a more engaging shopping experience and improves customer satisfaction.

BiLens offers e-commerce businesses a competitive advantage with its advanced search and recommendation features that surpass basic keyword matching. By integrating visual features and contextual captions, BiLens captures intricate visual patterns and interprets the contextual meaning of the captions. This approach enables a deeper understanding of user queries, resulting in more accurate product discovery and highly tailored recommendations, which enhances the precision of search results and better aligns product suggestions with user preferences. Moreover, BiLens features a flexible and adaptable architecture that scales across various product categories, allowing it to manage diverse product ranges and improve recommendations for a wide array of items in the e-commerce sector. This scalability ensures that e-commerce businesses can continuously integrate and utilize data from different categories, which enhances their ability to refine customer engagement strategies, improve service quality, and boost operational efficiency.

The growing demand for visual-based search on major e-commerce platforms and recent literature highlight the essential role of visual search technology in boosting customer satisfaction and the need for advanced recommendation systems (Abluton, 2022). By integrating AI with recommendation frameworks, BiLens enhances customer interactions and data processing accuracy, enabling businesses to offer personalized, dynamic service while addressing operational challenges.

Furthermore, this study's detailed evaluation of LLMs and word embeddings provides a benchmark for developing more effective recommendation frameworks. Researchers can build upon these benchmarks to improve model performance and flexibility. BiLens's scalability and adaptability underscore its value in delivering customized recommendations across various e-commerce domains, contributing to advancements in customer engagement and AI-driven service enhancement.

5.2. Contributions to literature and implications for practice

Existing literature predominantly explores visual-based search techniques (Dagan et al., 2021) and some multimodal approaches (Laenen et al., 2018). Our research extends these ideas by integrating visual-based features with LLM-generated captions, addressing the limitations of image-only methods and inaccuracies in keyword-based product descriptions. By fusing these elements, BiLens enhances visual search recommendations, offering a more comprehensive solution for overcoming traditional search method constraints.

Our work provides new insights into how large language models can improve search frameworks in e-commerce by leveraging both visual and textual data. This approach not only enhances the accuracy of search results but also resolves challenges associated with product titles and search precision. We focus on fully utilizing the user-uploaded image query to gather more information and context about the user's exact needs. Additionally, we shed light on how variations among different LLMs affect caption generation and subsequent search accuracy, offering a nuanced understanding of their practical applications. Our research also examines the impact of various word embedding techniques on search performance, contributing to more effective image search methodologies. These advancements set new benchmarks for evaluating multimodal retrieval systems, particularly using the Amazon product dataset. Our comparative analysis between text-only, image-only, and multimodal retrieval methods highlight the strengths and limitations of each approach, providing a comprehensive understanding of their respective contributions and effectiveness. These findings offer significant implications for refining search and recommendation systems, ultimately enhancing user experience and operational efficiency in e-commerce.

5.3. Limitations and future scope

BiLens has made significant strides in enhancing e-commerce recommendation systems through its integration of visual features and textual captions. However, there are notable areas for further development. Currently, the system operates with high-quality product images and their captions, which does not fully address the impact of image quality variability. Lower-quality images may affect processing and feature extraction, potentially influencing system performance. Additionally, while BiLens's emphasis on visual and textual data provides a solid foundation for generating accurate recommendations, incorporating other critical factors—such as brand, price, and product popularity—could further refine its capabilities. These elements are commonly utilized in e-commerce recommendation systems and could enhance the system's overall effectiveness. For demonstration purposes, the study focused on five randomly selected categories from the Amazon dataset. Extending the research to include a broader range of categories would offer a more comprehensive evaluation of BiLens's versatility and effectiveness across diverse product types.

Future research will address these limitations by integrating lower-resolution images, incorporating additional influential factors, and expanding the scope to cover a wider spectrum of product categories. These advancements will strengthen BiLens's capabilities, offering more precise and dynamic recommendations for e-commerce platforms and underscoring the ongoing need for innovation in recommendation technologies. This progression reflects the dynamic interplay between technological advancements and practical applications, setting a trajectory for future research and industry practice.

6. Conclusion

Aiming to advance visual search technology, this paper introduces BiLens, a multimodal recommendation framework that offers a novel and impactful approach for the e-commerce domain. By integrating both visual and textual features, BiLens addresses the limitations of traditional image-only or text-only search systems, enabling more accurate and context-aware product recommendations. Among the evaluated model combinations, the integration of Florence-2-large with BERT, alongside ViT for visual encoding, delivered the strongest performance, achieving an F1 score approximately 9 % higher than text-only methods and 2.88 times higher than image-only methods.

Importantly, statistical significance tests using the Wilcoxon signed-rank test confirmed that these improvements are not only substantial but also statistically meaningful ($p < 0.05$). These findings underscore the robustness of the BiLens framework and the effectiveness of multimodal fusion. Furthermore, incorporating visual representations of user queries and leveraging large language models for image captioning proved essential in enhancing recommendation quality across diverse product categories. The collective strengths of transformer-based architectures—Florence-2-large, BERT, and ViT—demonstrate their utility in effectively processing and aligning multimodal inputs, thereby setting a strong foundation for future advancements in e-commerce visual search and personalized product discovery.

CRediT authorship contribution statement

Anitha Balachandran: Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Mohammad Masum:** Writing – review & editing, Validation, Supervision, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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